

Saurabh Nair

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EDUCATION

University of Toronto

B.Sc Computer Science & B.A Economics [Double Major], CGPA- 3.73

September 2022 – June 2027

Awards: Dean's List Scholar (2022,2023,2024,2025), University of Toronto International Scholar Award (2022),

University of Toronto Scholar (2022)

EXPERIENCE

Software Developer Intern

May 2025 - Present

BAYER Canada

Toronto, ON

- Refactored and extended core components of the Radimetrics backend (Java, Spring Boot), reducing API response times by 28% and improving system scalability across multi-hospital deployments handling over 10,000 imaging records daily.
- Developed and integrated front-end features using React and TypeScript within the microservices architecture; contributed to a 35% decrease in UI-related support tickets by improving data reliability and feature tracking.

Machine Learning Intern

June 2024 - August 2024

Emirates Steel Arkan

Abu Dhabi, UAE

- Developed and implemented advanced machine learning models to optimize production processes and improve predictive maintenance, resulting in a 15% reduction in downtime and enhanced operational efficiency.
- Engineered a dashboard system using *Python-Dash* to monitor ferro alloy usage. Applied granular data analysis to construct a descriptive model, leading to an estimated reduction of \$1.2 million in operational costs for plant production

Back End Developer

May 2024– Present

Kaytoons Inc. [Startup]

San Jose, USA

- Engineered and maintained robust backend systems for *Kaytoons*, a startup based in San Jose, improving system reliability and scalability to support a growing user base, for a children's app that is going through funding.
- Implemented RESTful APIs and optimized database queries, enhancing application performance and enabling seamless integration with frontend services, and adding a *voice cloning feature using the PlayHT API system using Python*.

Teaching Assistant

September 2024 – Present

UTM Mathematical & Computational Sciences Department

Mississauga, ON

- Led tutorials and labs for CSC236: Theory of Computation, guiding students through advanced topics such as algorithm correctness, time complexity, and automata theory, enabling a strong grasp of theoretical computer science concepts.
- Oversaw grading and feedback for term work, quizzes, and exams, ensuring accuracy and timely responses for a large cohort, which improved overall student performance and engagement with course material.

PROJECTS

[MatherBot](#) | *React, CSS, Next.js, Material UI, OpenRouter API, Meta Llama AI*

- Developed *MatherBot*, an AI-driven math assistant chatbot, using *Next.js* for the frontend and *Material UI* for a sleek, dark-themed user interface, providing real-time support to students in solving math problems effectively.
- Integrated *OpenRouter API* to enable real-time, AI-powered responses within a responsive chat interface, optimized for performance and SEO with server-side rendering, and deployed on *Vercel* for reliable and fast delivery.

[Premier League Match Analysis and Prediction System](#) | *Python, BeautifulSoup, pandas, SciKit Learn*

- Developed a comprehensive data pipeline to scrape and process detailed match and shooting statistics for Premier League teams across multiple seasons using *Python, BeautifulSoup*, and *pandas*, with data sourced from the FBref website.
- Implemented a machine learning model using a *Random Forest classifier* to predict match outcomes based on historical data, aggregating and filtering data to focus on Premier League matches, and simulating entire seasons to generate final league tables, offering insights into team performance and potential season standings.

TECHNICAL SKILLS

Technical Languages: Java, Python, C, MySQL, JavaScript, R

Developer Tools: Git, Google Cloud Platform, PlayHT, OpenRouter, React, Node.js, Next.js, Firebase, Llama API

Libraries: NumPy, Matplotlib, SciKit Learn, Pandas, OpenCV, SpringBoot, Docker

Languages: English, Hindi (native), Malayalam (native), Arabic (certified diploma), French (novice).